IVIS19 Project 1: Visualize Student Group Formation based on Individual Skills and Interests

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Link :

http://rachhek.github.io/#

http://coon.nu/viz01/rachhek/

(alternative link)

Data Cleaning and Organization:

The dataset was converted to list of objects in JSON format. Null values in hobbies column were replaced by empty string. A new calculated column called skill_total(sum of all skills values) was added.


```
{
    "id": 5,
    "full_name": "Harry Potter ",
    "major": "Media Technology",
    "degree": "Master in Science",
    "hobbies": "",
    "learning_expectation": "Ledsen how to visualize heavy data in a intuitive way ",
    "infoviz_skill": 7,
    "stat_skill": 2,
    "maths_skill": 4,
    "draw_skill": 10,
    "computer_skill": 8,
    "programming_skill": 5,
    "cg_skill": 4,
    "hci_skill": 8,
    "ux_skill": 9,
    "communication_skill": 8,
    "collaboration_skill": 8,
    "repo_skill": 7,
    "skill_total": 80
},
```


Figure 1 Example of JSON representation

Technologies used:

D3.js, HTML, CSS, JQuery

Skill level Division

Expert : Skill level >=8

Intermediate : Skill level > =4

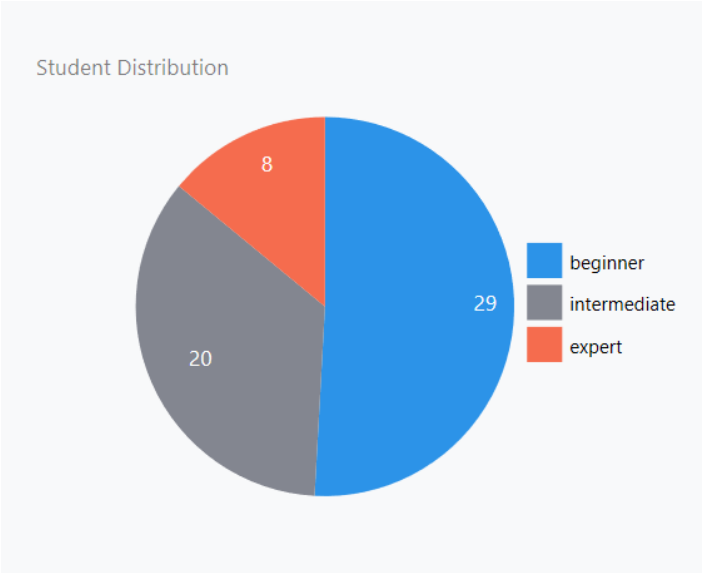
Beginner : Skill level <4

User Overall Division

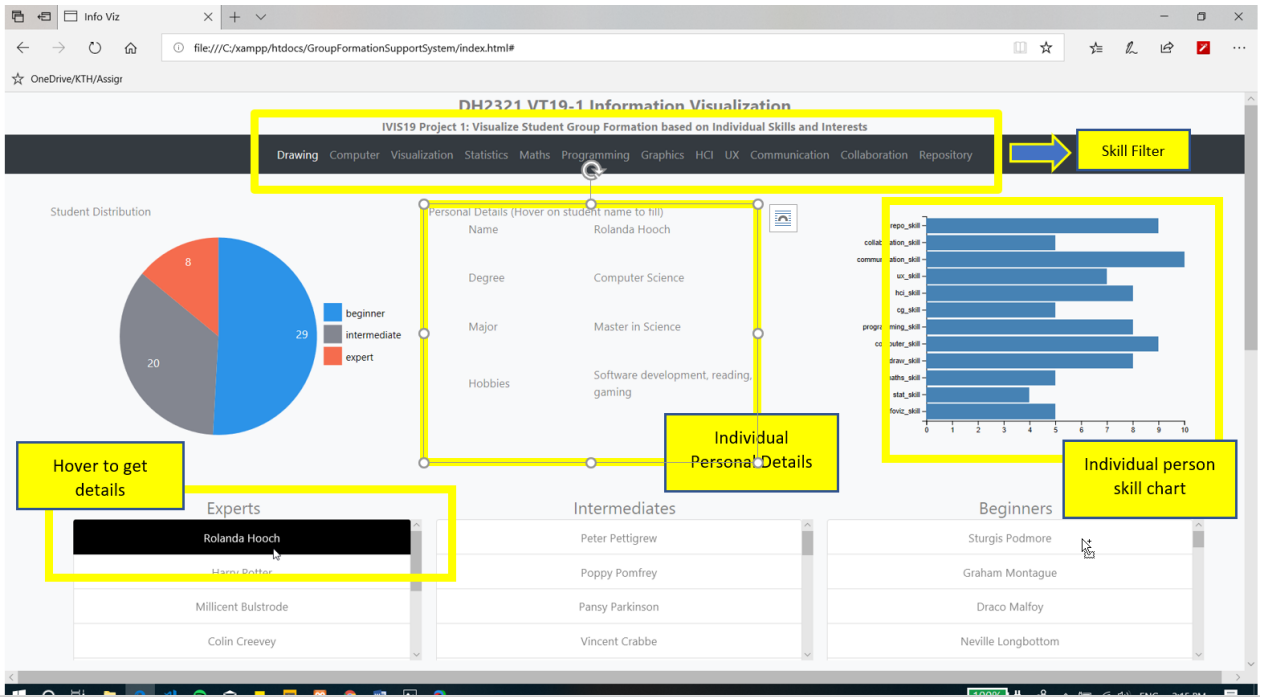
Overall Expert : Sum of Skills > 60% of total

Overall Intermediate: Sum of Skills >= 45% of total

Overall Beginner: Sum of Skills < 45% of total



Visualization for Decision Support



Logic for Group Selection:

A weighted sum of skills is calculated for every student. Users can set the weights for the skills using sliders on a range of 1 to 100. The list is sorted in descending order of weighted sum. A group of 5 is selected from the top of this list until everyone is assigned a group.

Figure 2 Sliders to set individual skill weight

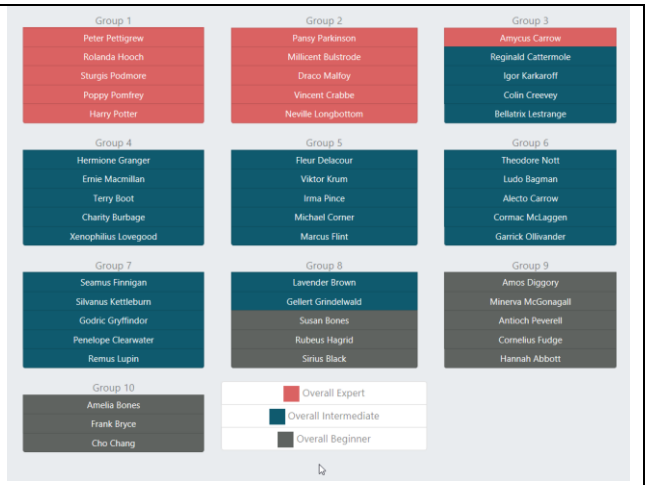


Figure 3 Division of groups based on weighted sum

Best Practices

"The most fundamental need in visualization is interactive selection of individual data entities or subsets of data entities. Users select entities to identify data that is of interest to them. This is useful for many reasons, including viewing detailed information about the entities…"[1]

[1] Chris, North. "Handbook of Human Factors and Ergonomics, Human–Computer Interaction, Chapter 43." In Visual Languages, 2012. Proceedings., IEEE Symposium on, pp. 1228. IEEE, 2012.